

Demo Board — Notes

Exported 2026-08-07 by PowerTree

Power Budget Sources

Power budget references

This vault collects **where every number came from** so the tree is auditable.

- Wall adapter rating: label on unit, 12 V / 2 A
- Regulator efficiencies: vendor datasheets (see child notes)

TPS54331 (5V Buck)

Linked elements: 5V Buck (Main 12V Rail)

TPS54331 efficiency

Datasheet Fig. 7: ~92 % at 12 Vin / 5 Vout / 1 A.

Vin	Iout	Eff
12 V	0.5 A	90 %
12 V	1 A	92 %

MCU current draw

Linked elements: MCU Icc (run) (Main 12V Rail), MCU analog Iq (Main 12V Rail)

STM32H7 Icc

DS: run mode @ 480 MHz, all peripherals: 120 mA typ, 180 mA max (85 °C).